

QingPeng Lam

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EDUCATION

University of Michigan

Bachelor of Science in Computer Science, Minor in UX Design | GPA: 3.8

Ann Arbor, MI

Aug. 2024 – Dec. 2026

EXPERIENCE

Software Engineering Intern

May 2026 – Present

USAA

San Antonio, TX

- Built 12 reusable C# AI skills and MCP integrations that automated production log retrieval, root-cause analysis, code-fix generation, unit testing, Git operations, and merge-request creation across 22 workflows and 6 internal projects.
- Investigated 120+ application and CI/CD issues by tracing logs, service interactions, source-code failures, and GitLab pipelines across containerized Podman/Docker environments.
- Developed search features across 6 Coveo Atomic applications, integrating internal and external content sources and implementing frontend, backend, and indexing workflows using JavaScript, Python, and C#.

Data Systems Engineering Intern

Aug. 2025 – May 2026

University of Michigan Institute for Social Research

Ann Arbor, MI

- Built Python and SQL data pipelines to clean, validate, deduplicate, and organize research datasets for downstream analysis and reporting.
- Developed REST APIs and reporting dashboards that converted complex research data into accessible metrics and tools for 200+ researchers and program stakeholders.
- Automated recurring research-support workflows, including data preparation, report generation, bibliography management, and documentation, reducing manual processing and improving consistency.

Software Engineer Intern

Aug. 2024 – June 2025

Michigan Hillel

Ann Arbor, MI

- Built Python, pandas, NumPy, and SQL ETL pipelines to consolidate, clean, and validate 3,000+ records from multiple operational data sources, reducing reporting errors by 35%.
- Developed JavaScript and Tableau dashboards with SQL data models to track program participation, student engagement, and operational performance for planning and reporting.
- Integrated third-party APIs with centralized SQL workflows, replacing disconnected data processes and improving the completeness and consistency of recurring reports.

PROJECTS

DevDoctor | Python, FastAPI, PostgreSQL, Ray Serve, MLflow, Docker

June 2026 – Present

- Architected and built a multi-tenant platform that converts Sentry, GitHub, and GitLab failures into clustered incidents, AI-generated diagnoses, sandbox-validated patches, human approvals, and automated Draft PRs/MRs.
- Productionized the ML lifecycle with MLflow model versioning and evaluation, Ray Serve autoscaling and monitored inference, PostgreSQL row-level security, OIDC/SCIM access control, durable workers, encrypted recovery, and 260+ automated tests.

Immich AI Photo Search | Node.js, Express, Transformers.js, Vector Search

Feb. 2026 – May 2026

- Architected a full-stack AI photo intelligence platform that transforms uploaded media into a searchable knowledge base using vision-generated captions, semantic embeddings, EXIF metadata, and hybrid vector-keyword retrieval.
- Engineered a fault-tolerant asynchronous ingestion pipeline with automatic HEIC conversion, resumable AI indexing, graceful model fallbacks, and SHA-256 idempotency controls, recovering all 14 failed assets in a 19-image compatibility test.

TECHNICAL SKILLS

Languages: Python, C#, Java, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, React, Node.js, Express, Ray Serve, MLflow, Transformers.js, PyTorch, TensorFlow, pandas, NumPy

Data & AI: PostgreSQL, Vector Search, Embeddings, AI Agents, Model Serving, ETL, Tableau, MCP, Generative AI

Tools & Infrastructure: Docker, Podman, Git, GitHub, GitLab CI/CD, Linux, MCP, REST APIs, Unit Testing

Spoken Languages: Mandarin, Cantonese, English